



Stats & Stumps: A Novel Impact Factor for Player Comparison and Match Prediction in Cricket

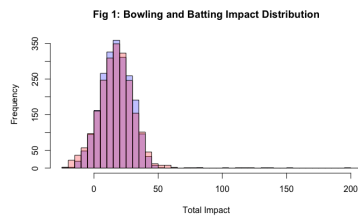
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BACKGROUND

- Cricket is **the world's second most popular sport**, second only to football
- The new T20 format has exploded in popularity for its entertainment value and **higher outcome variance**
- New players/styles emerged, as well as a **betting market valued at \$14.5bn**.
- The goal is:
 - Determine a number (impact factor) that allows the impact of bowlers to be compared to batters quantitatively**
 - Use that number to predict future matches**
 - Develop a web app to simulate and visualize these comparisons**
- This will enable selectors to make informed decisions based on past play and compare different players across classes (batsmen, bowlers, all-rounders)

DATA

- ESPN Cricinfo**
 - Used the Statsguru database to select **1,061 T20Is over 20 years** between full ICC member nations
 - Aggregated the player data from every match
 - Computed the impact factor for batting and bowling
 - ~50,000 player observations**



- Impact factor**
 - 2132 observations, a batting and bowling impact for every team in every match
 - Formula was determined by optimizing the distribution similarity between team batting and bowling impacts (found in methods)
 - KS Test performed to verify similarity ($p = 0.37$)
- Predictor and Response variables**
 - Attempting to predict a win probability through player batting/bowling impacts
 - Make venue considerations based on where it's easier to bat or bowl
 - Assume the top 8 rated batsmen in the Playing XI are meant to bat, and the top 6 rated bowlers in the Playing XI are meant to bowl
 - 14 players per team * 2 teams = 28 player-level features to use in match prediction

METHODS

- Batting Impact** = $(R \times 0.125) + ((SR - 130) \times 0.025) + (4s \times 0.3) + (6s \times 0.5) - \text{OutPenalty}$
 - R is runs scored
 - SR is Strike Rate (runs/ball)
 - $4s$ and $6s$ are the respective boundaries
 - OutPenalty is a fixed value of 0.5 if a batsman gets out
- Bowling Impact** = $(W \times 3.25) + (M \times 3) - ((E - 7.4) \times 0.5)$
 - W is wickets
 - M is maiden overs
 - E is economy
- Total impact = Bat impact + Bowl impact for individual all-round performance
- Formula determined by optimizing for match bat/bowl impact similarity (KS-Test)
- Built 3 models to try and predict matches based on average player batting and bowling impact factor per game
 - Naïve baseline: Higher total team impact wins**
 - Logistic regression w/ regularization (LR)**
 - Random forest classifier (RF)**
 - Support Vector Machine (SVM)**
 - Trained with 5-fold internal CV for hyperparameter optimization
- Scaled the player impacts based on venue-level impacts (if a ground favored batters or bowlers)
- Build **web app using Shiny** to display results

RESULTS

Impact Factor Leaderboards

Rank	Player	Career Impact	Player	Impact per Game
1	V Kohli (IND)	608.06	Abhishek Sharma (IND)	6.94
2	JC Buttler (ENG)	603.5	PD Salt (ENG)	5.48
3	RG Sharma (IND)	599.65	NT Tilak Varma (IND)	5.43
4	Imran Tahir (SA)	530.37	SA Yadav (IND)	5.13
5	MJ Guptill (NZL)	511.31	V Kohli (IND)	5.11

Table 1: Batting Impact Top 5: Career & Per Game

Rank	Player	Career Impact	Player	Impact per Game
1	Rashid Khan (AFG)	525.49	Rashid Khan (AFG)	6.22
2	TG Southee (NZL)	448.07	Rashid Khan (AFG)	6.11
3	AU Rashid (ENG)	427.1	BAW Mendis (SL)	5.95
4	Shakib Al Hasan (BAN)	421.38	CV Varun (IND)	5.87
5	PW Hasaranga (SL)	404.73	Imran Tahir (RSA)	5.74

Table 2: Bowling Impact Top 5: Career & Per Game

Rank	Player	Total Impact	Player	Impact per Game
1	Shakib Al Hasan (BAN)	658.5	Abhishek Sharma (IND)	7.81
2	Rashid Khan (AFG)	592.75	Rashid Khan (AFG)	6.89
3	HR Pandya (IND)	556.42	SR Watson (AUS)	6.23
4	GJ Maxwell (AUS)	548.05	Shakib Al Hasan (BAN)	5.88
5	Shahid Afridi (PAK)	527.63	PW Hasaranga (SL)	5.83

Table 3: All-Rounder Impact Top 5: Career & Per Game

Rank	Top Bowling Grounds	Top Batting Grounds
1	Dallas	Gwalhati
2	New York	Hyderabad
3	Kingstown	Blomfontein
4	Colombo (PSS)	Brasourne
5	Port of Spain	Mohali

Table 4: Top 5 Batting/Bowling Venues

Best Fit Model: ElasticNet, 72% accuracy ($\alpha = 0.1, \lambda = 0.055$)

Model	Accuracy	RMSE	Log-Loss	Composite Score	95% Confidence Interval
Heuristic	0.6598	NA	NA	NA	NA
Logistic Regression	0.7170	0.4474	0.5876	1.3181	(1.2997, 1.3364)
Random Forest	0.6840	0.4519	0.5959	1.3639	(1.3391, 1.3887)
SVM	0.7028	0.4441	0.5813	1.3226	(1.3017, 1.3434)

Note: Composite Score is calculated as $(1 - \text{Accuracy}) + \text{RMSE} + \text{Log-Loss}$. Lower values indicate better overall model performance.

Table 5: Model performance based on mean accuracy and 95% confidence intervals over 30 iterations

Shiny App Visualizations – Match Simulator

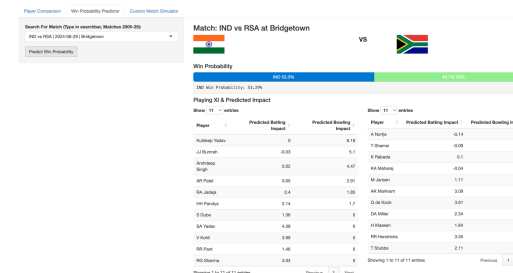


Fig 2: Backtested match: IND vs. SA T20WC final

Shiny App Visualizations – Player Comparison

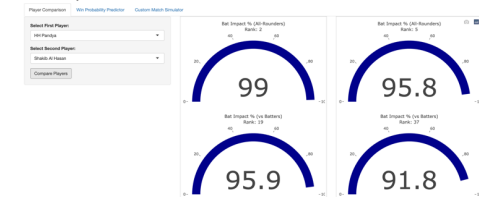


Fig 3: Player Comparison Tool

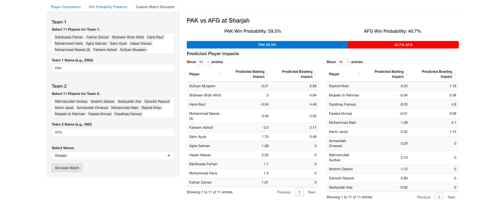


Fig 4: Predict future match by Playing XI

CONCLUSIONS/QR CODES

- Impact formulas create similarity between bat/bowl data
- 3 models were fit, compared to baseline
 - LR with regularization was the best model with **~72% test accuracy**
 - Minimized composite score accounting for log-loss and RMSE
- Shiny App provided good visual & easy future tests: model **correctly predicted 90% of recent Asia Cup matches**
- Next steps: Refine impact factor methodology, translate impact factor difference into margin of victory

ShinyApp:
shorturl.at/MmUoE



GitBook code:
shorturl.at/eu3gh



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